PROJECT REPORT

SMART IRRIGATION SYSTEM WITH TEMPERATURE MONITORING

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CLASS: IOT-B

AIM:

To design and implement a smart irrigation system that monitors soil moisture levels and ambient temperature, displaying real-time data on an LCD and controlling irrigation via a servo motor.

COMPONENTS REQUIRED:

SNO	COMPONENT NAME	NOS
1	Arduino UNO	**1**
2	Bread board	**1**
3	LCD display	**1**
4	Soil moisture sensor	**1**
5	Temperature sensor	**1**
6	Servo motor	**1**

METHODOLOGY:

CONNECTIONS:

1)Temperature Sensor (LM35): Connect the **VCC** pin to **5V**, the **GND** pin to **GND**, and the **OUT** pin to **A2** (Analog input pin).

2)**Soil Moisture Sensor**: Connect the **VCC** pin to **5V**, the **GND** pin to **GND**, and the **OUT** pin to **A0** (Analog input pin).

3)**Servo Motor**: Connect the **VCC** pin to **5V**, the **GND** pin to **GND**, and the **Control (signal)** pin to **Pin 9** (PWM pin).

5) **LCD Display**: Connect **VCC** to **5V**, **GND** to **GND**, **SDA** to **A4**, and **SCL** to **A5**.

PROCEDURE:

1)**Setup Phase**: Initialize the serial monitor, LCD, and servo motor in the setup () function. We have loaded the necessary header files and the lcd and everything is set up

2)**Temperature Read**: The analogue input from the LM35 temperature sensor is read, converted to voltage, and then to temperature (°C).the temperature sensor is used to read the values.

3) **Soil Moisture Read**: Read the soil moisture sensor's analogue value and map it to a percentage (0-100%).

4)**LCD Display**: Display the temperature and soil moisture readings on the LCD screen.

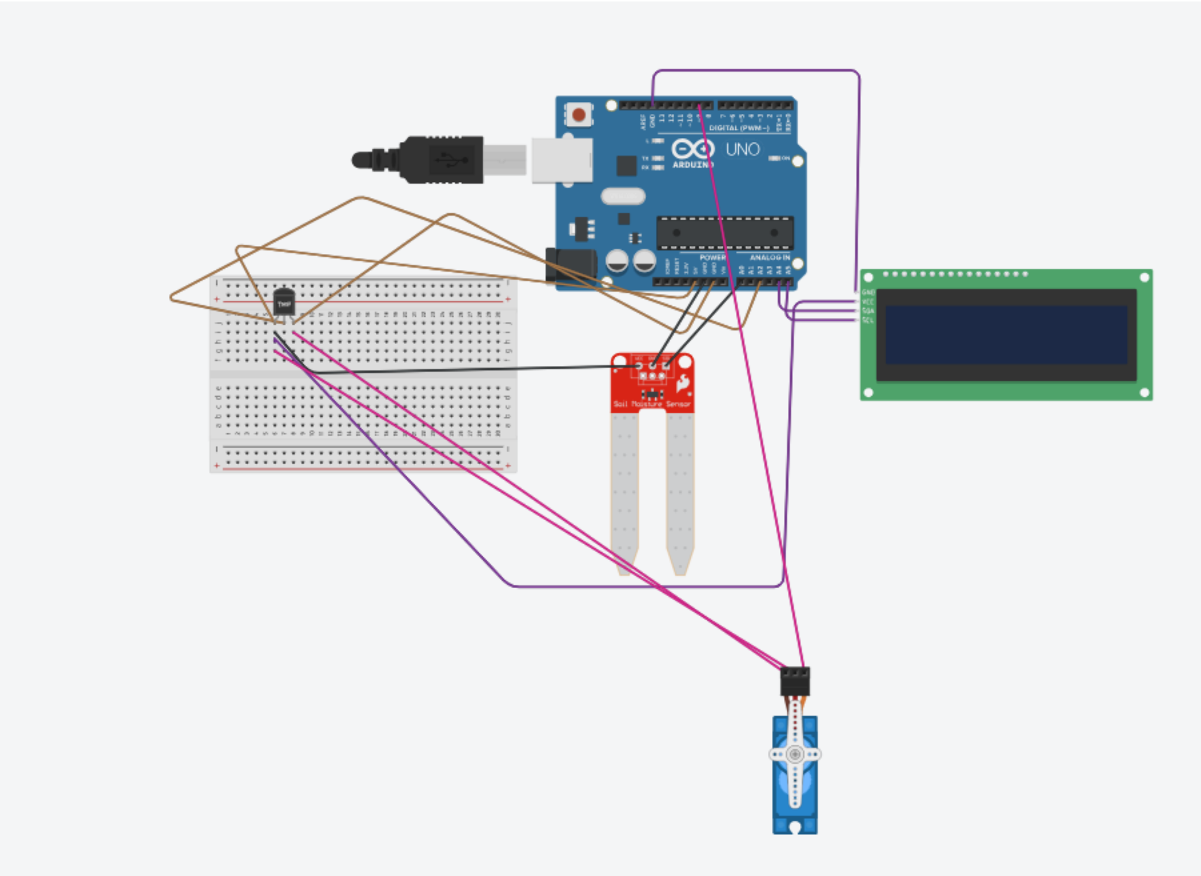
5) **Servo Control**: If the soil moisture is below 40%, the servo is turned on (90°), indicating low moisture. When the soil moisture is below 40% then the servo motor which acts as a tap is turned on to restore the moisture.

7) **Servo Off**: If soil moisture is above 40%, the servo is turned off (0°).it is turned off when the moisture is sufficiently above 40%

8) **LCD Update**: Continuously update the LCD with the status of the servo motor (ON/OFF).

9) **Repeat**: The process repeats every 2 seconds to update the sensor readings and servo state.

CIRCUIT DIAGRAM:



CODE:

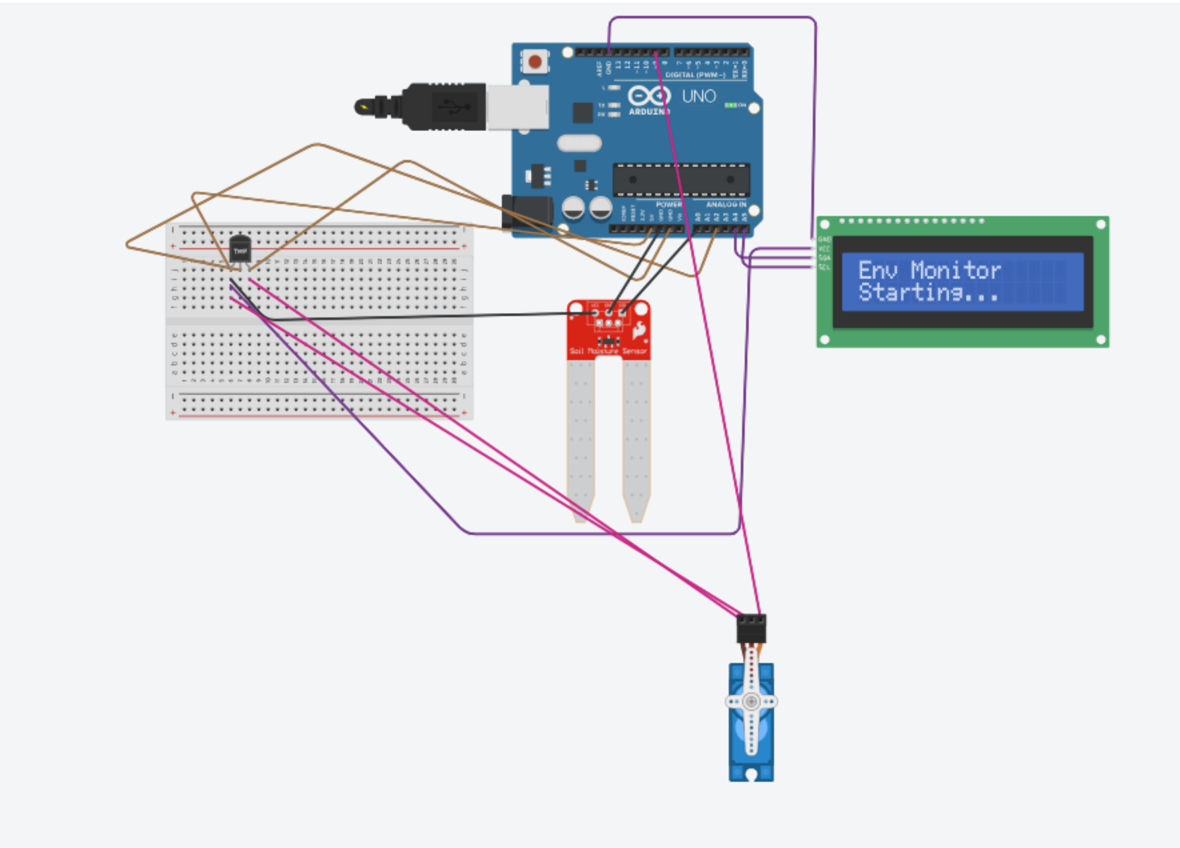
```
#include <Wire.h>

#include <LiquidCrystal_I2C.h>

#include <Servo.h>
```

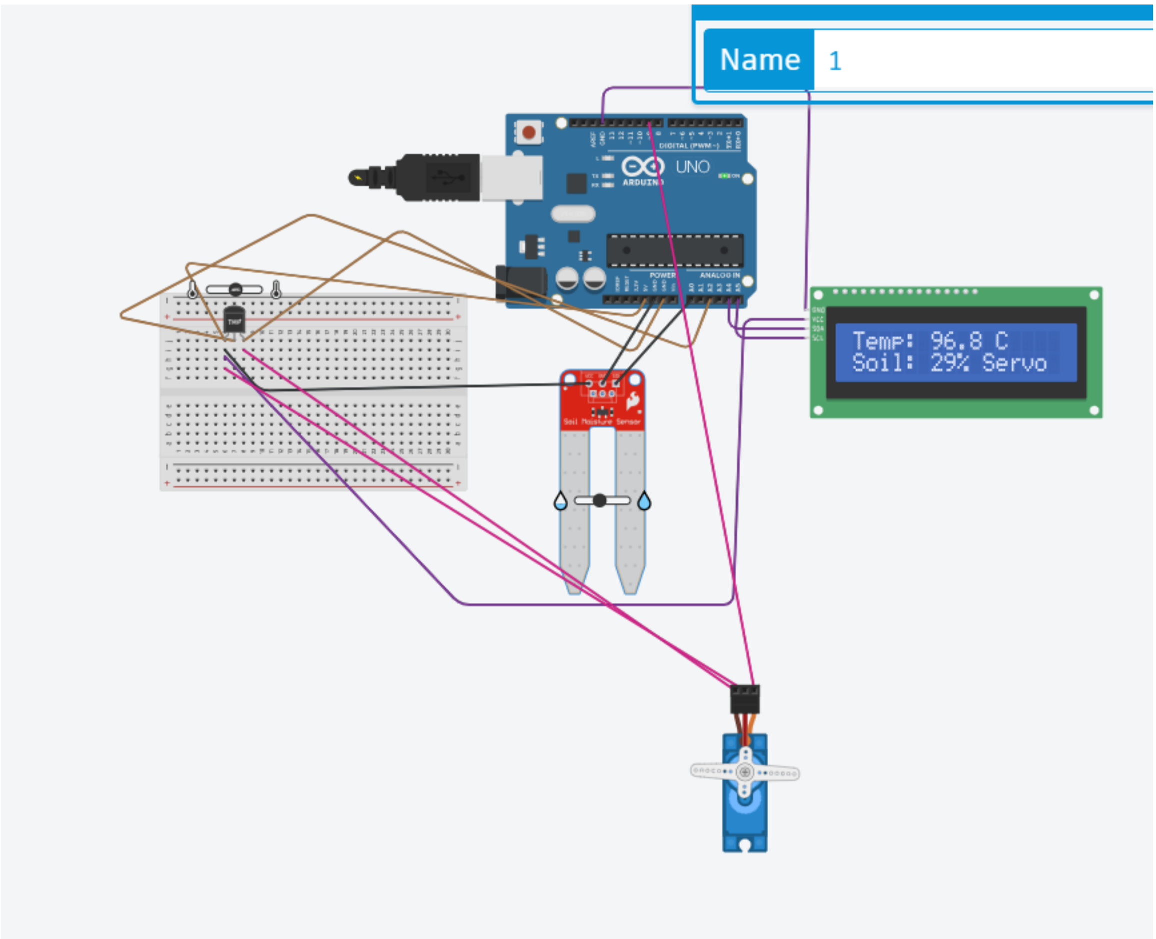
```
#define TEMP_SENSOR_PIN A2
#define SOIL_SENSOR_PIN A0
#define SERVO_PIN 9


LiquidCrystal_I2C lcd(0x27, 16, 2);


// Create a Servo object
Servo servo;

void setup() {
  // Initialize Serial Monitor for debugging
  Serial.begin(9600);

  // Initialize LCD
  lcd.init();
  lcd.backlight(); // Turn on the backlight

  // Display a startup message
  lcd.setCursor(0, 0);
  lcd.print("Env Monitor");
  lcd.setCursor(0, 1);
  lcd.print("Starting...");
  delay(2000); // Wait 2 seconds
  lcd.clear();

  // Initialize the servo motor
  servo.attach(SERVO_PIN);
  servo.write(0);
```

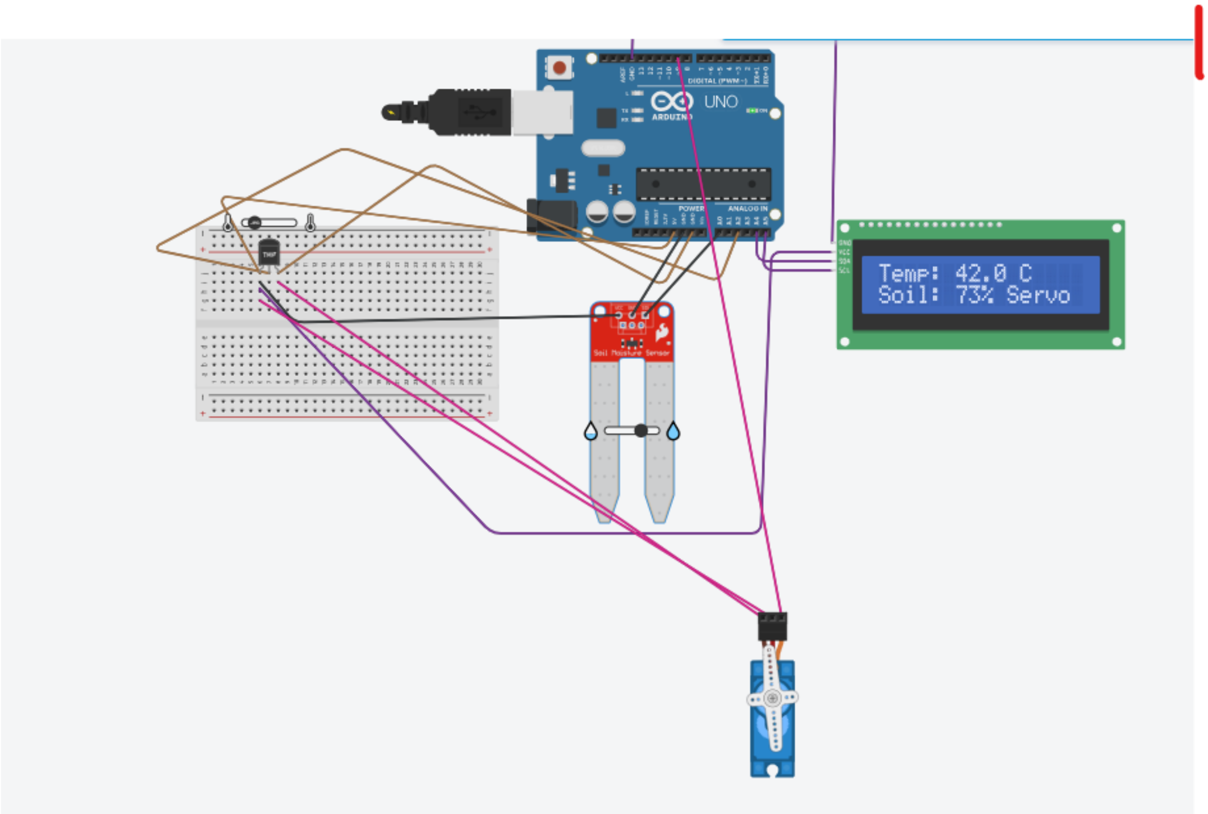
```
}

void loop() {
 // Read temperature sensor value
 int tempSensorValue = analogRead(TEMP_SENSOR_PIN);
 float voltage = tempSensorValue * (5.0 / 1023.0);
 float temperature = voltage * 100.0;



 if (temperature < 0) temperature = 0;
 if (temperature > 100) temperature = 100;

 // Read soil moisture sensor value
 int soilSensorValue = analogRead(SOIL_SENSOR_PIN);
 int soilMoisturePercent = map(soilSensorValue, 0, 1023, 0, 100);

 // Display readings on the LCD
 lcd.setCursor(0, 0);
 lcd.print("Temp: ");
 lcd.print(temperature, 1);
 lcd.print(" C");

 lcd.setCursor(0, 1);
 lcd.print("Soil: ");
 lcd.print(soilMoisturePercent);
 lcd.print("%");

 // Servo control logic based on soil moisture
 if (soilMoisturePercent < 40) {
  servo.write(90);
```

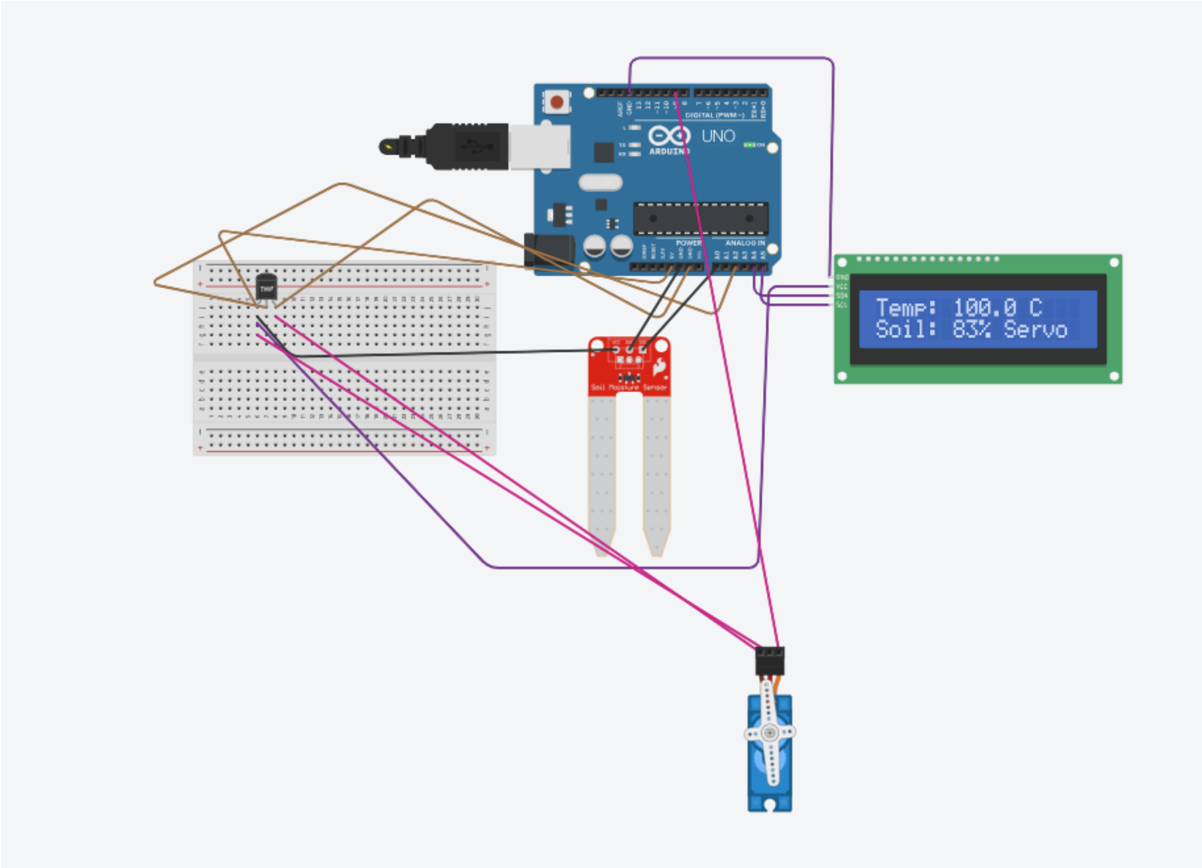
```
   lcd.setCursor(10, 1);
   lcd.print("Servo ON");
  } else {
   servo.write(0);
   lcd.setCursor(10, 1);
   lcd.print("Servo OFF");
  }

  delay(2000);
}
```

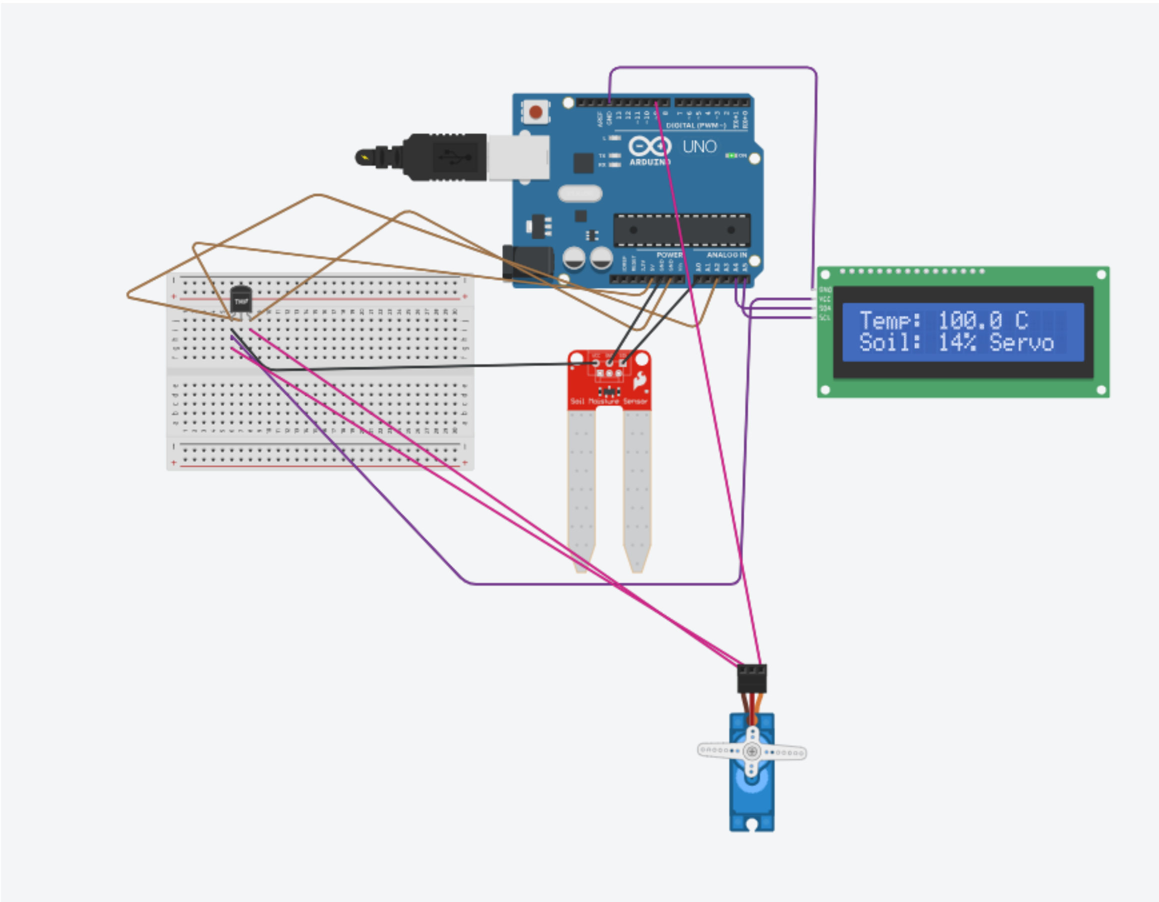
OUTPUT:




Name
1
UNO
ARDUINO
Temp: 96.8 C
Soil: 29% Servo
Soil Moisture Sensor


UNO
ARDUINO
Temp: 42.0 C
Soil: 73% Servo
Soil Moisture Sensor


UNO
ARDUINO
Temp: 100.0 C
Soil: 83% Servo


UNO
ARDUINO
Temp: 100.0 C
Soil: 14% Servo

RESULTS:

Hence the system built successfully monitors soil moisture levels and the changing temperature. The soil moisture and the and the servo motor have been established successfully to adjust according to the needs.